The Dirac Delta "Function"

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February 2024

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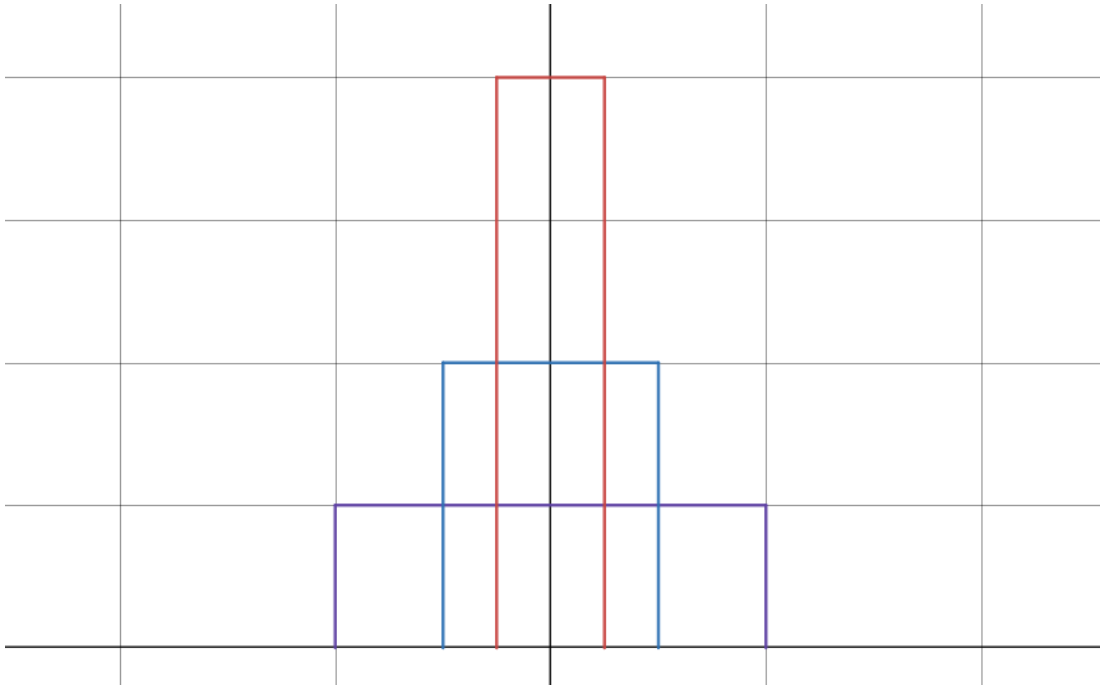

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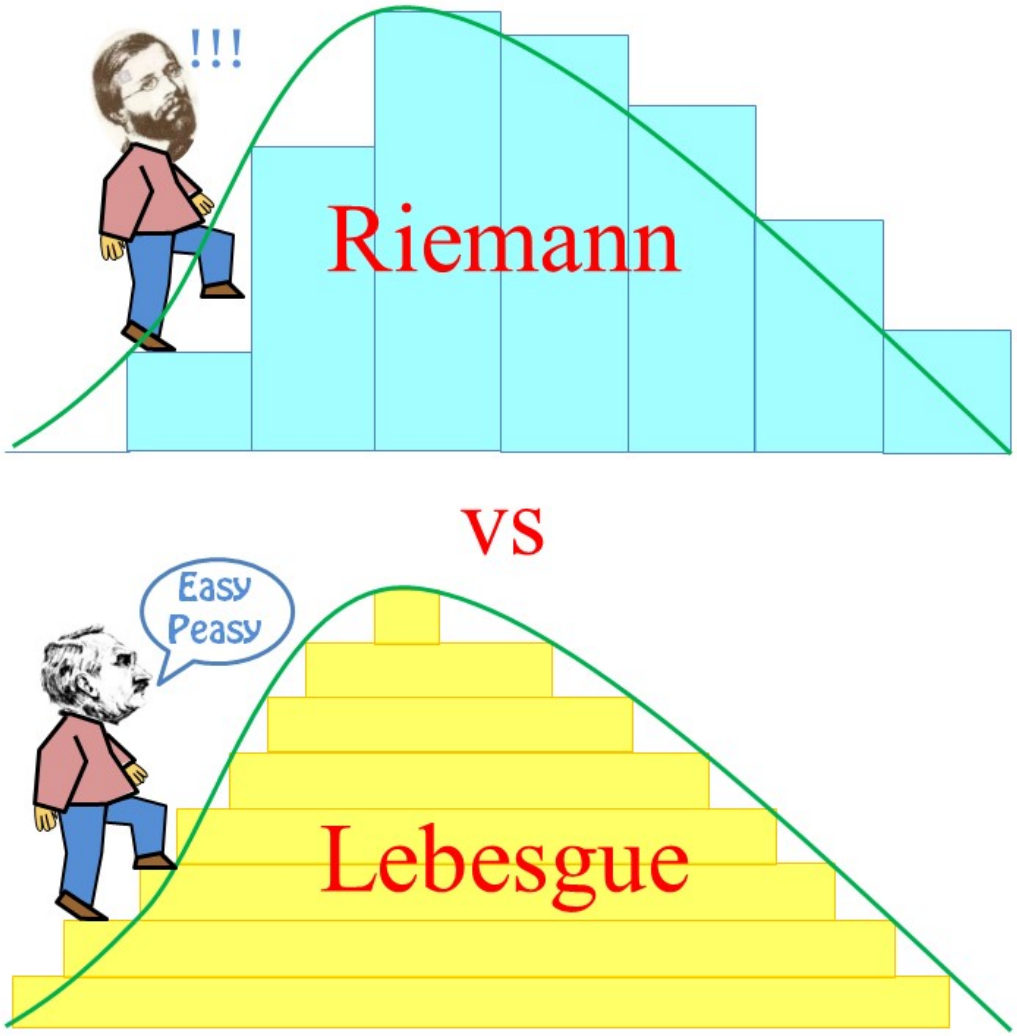
Riemann vs. Lebesgue Integral



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- L^2 forms a metric space defined by:

$$d(f, g) = ||f - g||$$

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$$= n - m \not\to 0$$

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$$\delta_0 \to \infty \text{ as } \sigma \to 0 \text{ so } ||\delta_0|| = \infty$$

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- δ is not a function so it does not make sense to integrate over
- Even if it were a function, the function definition is not continuous

$$\int \delta(x)\delta(x)dx = \int \delta(x)d\delta$$